Water Ledger Virginia: methods brief

Model version 1.11 · data snapshot 2026-11-09 · waterledger.org · github.com/northlinehydrology/waterledger

What it estimates

For any city, town, village, township or county in Virginia, Water Ledger estimates the water a data center campus of a given size and cooling design would withdraw, consume and return on an average day and on a 1-in-10-year peak day, and compares it with the local water system, sewer plant, downstream waters and electric utility. All inputs are public data.

Data

Category	Source	In Virginia
Places	Census TIGERweb: incorporated places; county subdivisions that are governments; counties	322 places: 227 cities, towns and villages; 95 counties
Water supply	EPA community water system service areas, with SDWIS population and connections	903 systems; 89% of places have one mapped
Wastewater	EPA Clean Watersheds Needs Survey 2022 (design flow) matched to ECHO (recent flow, receiving water)	238 plants; 99.7% of places have one within 40 km
Receiving waters	USGS Watershed Boundary Dataset; EPA ATTAINS (Virginia DEQ, 2024 report)	271 watersheds with a downstream path; 770 watersheds on those paths
Climate	Copernicus ERA5, 1995-2024, at each place's 0.25-degree cell (the nearest at least half land), corrected toward nearby NOAA and Environment and Climate Change Canada stations; NOAA stations by climate division for rainfall and road salt; Atlas 14 design storms; Climate at a Glance rainfall	148 cells, 147 corrected toward stations; 6 climate divisions
Energy	HIFLD retail service territories (PNNL copy); EIA-861 sales and peaks; EIA-923 plant reports for power-plant water	36 utilities with loads; grid regions: CPLE, DUK, PJM, TVA

Overall method (all states)

Weather (model 1.9). Each place takes its hourly temperature and dew point, 1995-2024, from its own ERA5 grid cell (0.25 degrees, about 10 km away), or the nearest cell at least half land where its own is mostly water. Its weather is corrected month by month and by time of day toward up to four weather stations within 100 km: where the cell is under 90% land, stations near the shore, since ERA5 runs cool beside lakes and the sea; elsewhere, stations inland. Hybrid and assist estimates depend most on the hottest hours, and so on this.

Cooling water. Hourly temperature and humidity drive the heat rejected by the campus under four cooling designs: cooling towers; hybrid, evaporative in warm weather only; dry coolers with adiabatic pre-cooling above a switch temperature (20-30 C, assumed); and closed loop with no evaporative cooling. Evaporation is the heat load times the evaporative fraction of Rutberg et al. (2011), as used in the USGS cooling tower method. Blowdown is evaporation divided by cycles of concentration minus one (DOE FEMP).

Uncertainty. Each estimate is 400 Monte Carlo runs, sampling 17 cooling parameters from published or estimated ranges; estimated ranges are marked [A]. We report the 10th, 50th and 90th percentiles. The peak day is the 1-in-10-year maximum day, taken as the 90th percentile of the annual maximum daily withdrawal.

Other water. Staff use, 10-30 gallons a person a day, scaled to the number of people on site; maintenance and cleaning, up to 2,000 gallons a day plus 2-20 per MW, bracketed by applicants' projections; and make-up water for closed loops, which lose 1-5% of their volume a year to leaks.

Source and wastewater. If a community water system is mapped in the place, the campus is served by it. If not, a closed loop connects to the nearest system serving more than 3,300 people when it is within 10 km; all other cases use new wells. Discharge goes to the nearest sewer plant if it is within reach and, for evaporative designs, has enough unused capacity for the blowdown. Otherwise closed loops use an on-site septic system and evaporative designs discharge directly.

Downstream. The place's main HUC-12 watershed is traced downstream for up to 25 watersheds, across state lines where needed, with the EPA-listed impairments and TMDLs for each.

Electricity use. Worked out hour by hour from the same weather and heat balance as the cooling water (after Lei and Masanet 2022): computers, power losses, fans, pumps, and chillers when outside air is too warm. Some inputs are assumptions, marked [A], including how much of a campus is liquid-cooled.

Electricity. Utilities come from the national retail territory layer, ordered by number of customers because it draws many small utilities as whole counties. Power-plant water is each state's own: 2024 plant reports to EIA, with Macknick et al. (2012) factors where a report is implausible or missing, weighted for imports, updated with 2022 reports and Siddik et al. (2024) on the same

basis. States can add their own layers (reported water use, wells, coldwater streams, state rules) on top of this method.

What is specific to Virginia

Virginia follows the overall method with national data throughout. What is particular to it:

Weather. 148 ERA5 cells; 147 cells corrected toward 32 stations: Andrews Air Force Base Airport, Baltimore-Washington Intl Airport, Charlottesville Albemarle Arpt, Danville Regional Airport, Davison Aaf Airport, Elkins-Rand Co-J.Rand Field Airport, Etrn Wv Rgnl/Shpherd Fd Ap, Hgrtwn Rgnl-R.Hnson Fld Ap, Julian Carroll Airport, Langley Afb Airport, Ldon-Crbn Apt-Mgee Fld Apt, Lynchburg Rgnl/Preston Glenn Fld Ap, Mercer County Airport, Naval Air Station, Norfolk International Airport, Norfolk Nas, Nwpt News/Wimburg Intl Apt, Oceana Nas, Piedmont Triad International Airport, Quantico Mcaf, Raleigh County Memorial Airport, Raleigh-Durham International Ap, Richmond International Airport, Roanoke Regional/Woodrum Field Ap, Rocky Mount-Wilson Rgn Apt, Ronald Reagan Washington Natl Ap, Salbry-Ocn Cty Wico Rgnl Ap, Smith Reynolds Airport, Sussex County Airport, Tri-Cities Regional Tn/Va Airport, Wallops Flight Facility Airport, Washington Dulles International Ap. Rainfall and road salt come from 6 NOAA climate divisions, with these stations: Richmond International Airport (2 divisions); Lynchburg Rgnl/Preston Glenn Fld Ap; Washington Dulles International Ap; Roanoke Regional/Woodrum Field Ap; Raleigh County Memorial Airport.

Places. 227 cities, towns and villages; 95 counties. Counties are included so that unincorporated land is covered.

Water quality. Impairments and TMDLs from Virginia DEQ's 2024 integrated report, through EPA ATTAINS.

Not yet added: Virginia's reported water use by sector, domestic wells, coldwater streams, lake-source cooling and state rules.

Example: 300 MW of IT capacity in Loudoun County

Million gallons per day, median with the 10th-90th percentile range. Peak day is the 1-in-10-year maximum day. Consumed water is evaporated or otherwise not returned. Source and wastewater are the defaults described above (muni: community water system; sewer: nearest sewer plant).

Design	Withdrawn, average day	Withdrawn, peak day	Consumed, average day	Source / wastewater
Cooling towers	2.07 (1.66-2.57)	2.92 (2.36-3.59)	1.64 (1.29-1.96)	muni / sewer
Hybrid	0.24 (0.19-0.30)	2.23 (1.80-2.77)	0.19 (0.15-0.23)	muni / sewer
Closed loop with assist	0.29 (0.23-0.40)	2.29 (1.78-2.95)	0.23 (0.18-0.30)	muni / sewer
Closed loop	0.01 (<0.01-0.01)	0.01 (<0.01-0.01)	<0.01	muni / sewer

Checks on this snapshot

Coverage of Virginia's places: water system 89%, watersheds 100%, utilities 98%, climate division 99.7%, a sewer plant within 40 km 99.7%. 50 randomly selected places were run through every step and cooling design with 0 errors. Most places without a mapped water system are rural areas on private wells.

Limits

These are screening estimates from public data, intended for comparing designs and sizes before an applicant's figures are available. They are not a substitute for facility design data, a utility capacity analysis or a permit review. Downstream paths are traced between HUC-12 watersheds, not along individual stream reaches. EPA assessments are from each state's most recent integrated report. Weather covers 1995-2024.

Planning-level hydrologic assessment. Not engineering design. Built with the open waterledger Python packages (waterledger 3.15.0); methods, parameters and sources: waterledger.org and the package repository. Northline Hydrology.